Vali Hameed

Valihameed88@gmail.com | 07305319083 | Vali-Hameed.com | linkedin.com/in/Vali-Hameed | github.com/Vali-Hameed

Education

Lancaster University September 2024 – June 2027

- Computer Science BSc with Honours (Year 2) - Expected First Class
- First Year Average - 70%

Experience

Co-Founder & Lead Engineer, Picky Eater – Lancaster, ENG October 2025 – June 2026

- Co-founded and led a 7-person team as the **top contributor (60% of commits)**, architecting a full-stack platform across web (**React/Vite**), mobile (**React Native**), and backend (**Django REST**).
- Designed a **polyglot data architecture** using **MongoDB** for social content and **PostgreSQL** for relational data, balancing NoSQL flexibility with strict **ACID guarantees**.
- Built a **7-signal recommendation algorithm** using order history, recency decay, dietary matching, and MongoDB aggregations, achieving **O(n log k)** selection performance.
- Engineered an **API security framework** (JWT auth, rate limiting, input sanitization) and authored regression test suites to harden against **OWASP Top 10** vulnerabilities.
- Implemented **real-time WebSocket chat** via **Django Channels/Redis**, and established PR workflows and CI/CD pipelines to increase milestone completion by **15%.**

Software Engineering Intern, DigbySwift – Leeds, ENG June 2023

- Developed a **cross-platform mobile application** from concept to completion using **Dart** and **Flutter**, delivering a user-centric movie and TV series search tool and **reduced** app search time by **25%**.
- Ensured code reliability and software quality by implementing comprehensive **unit tests**.
- Managed project version control using **Git** and **GitHub**, employing a feature-branching workflow with pull requests for code review before merging into the main branch.

Projects

Orbital Risk

- **Architected a mission control dashboard** using **Next.js 14** (a **React** framework) and **TypeScript**.
- **Engineered a real-time 3D visualisation** system to map satellite orbits and debris fields, using **WebGL** and **Three.js**. Enabling interactive trajectory analysis for safe launch planning.
- **Implemented a custom risk assessment engine** that calculates collision probabilities by processing orbital parameters (altitude, inclination, corridor width, time of launch) against real-time debris density datasets.
- **Developed a Python-based weather prediction service** to analyse atmospheric conditions and generate automated Go/No-Go launch recommendations through **FastAPI** endpoints and an **XGBoost** model.

UFC Fight Predictor

- Developed a **machine learning** model in **Python** to predict UFC fight outcomes by cleaning and processing a dataset of over **6,000** fights with **Pandas** before training a **Logistic Regression** classifier with **66%** accuracy.
- Deployed a **model-serving microservice** to **AWS ECS** by containerising a **FastAPI** backend with **Docker**.
- **Engineered** the core **web** backend by developing a secure **RESTful** API using **Java Spring Boot** and **Spring Security**, managing user data with **Spring Data JPA** and a containerised **PostgreSQL** database using **Docker**.

Technical Skills

Languages: Java, Python, Dart, JavaScript, TypeScript, SQL, C, C++, Haskell

Frameworks/Libraries: Flutter, Next.js, Tailwind CSS, React, Pandas, Scikit-learn, FastAPI, Spring Boot, Django

Tools & platforms: Git/GitHub, Unit Testing, Docker, AWS, PostgreSQL, MongoDB, MySQL, Vercel, Render